Infrarenal Abdominal Aortic Aneurysm

Simon De Freitas, MD, Nicole D'Ambrosio, MD,
Javairiah Fatima, MD*

KEYWORDS

• Abdominal aortic aneurysm • Endovascular • Open repair

KEY POINTS

- Endovascular abdominal aortic aneurysm repair has replaced open surgery as the primary modality for aneurysm repair.
- Screening programs are effective in reducing aneurysm-associated mortality in selected populations.
- Rutuptured aneurysms are are almost uniformaly fatal, irrespecive of the type of repair used and mortality reduction is acheived by repair in the asymptomatic stage.

INTRODUCTION

Abdominal aortic aneurysms are found in up to 6% of men and 1.7% of women over the age of 65 years and are usually asymptomatic.[1,2] The natural history of aortic aneurysms is continued dilation leading to rupture, which is associated with an overall 80% mortality. Of the patients with ruptured aneurysms that undergo intervention, half will not survive their hospitalization.[3] Reduction in aneurysm mortality is therefore achieved by prophylactic repair during the asymptomatic period. On a population-based level, this is supported by abdominal aortic aneurysm screening programs. Approximately 60% of abdominal aortic aneurysms are confined to the infrarenal portion of the aorta and are amenable to repair with off-the-shelf endovascular devices. Endovascular techniques have now replaced open surgery as the primary modality for aneurysm repair.[4] Several devices are available, and each has unique features that the operator must be familiar with when selecting the most suitable device. One major limitation of endovascular repair is the development of endoleaks, which may require reintervention over time. Currently in the United States, over 80% of infrarenal aneurysms are treated with endovascular techniques.[5] Open

Department of Vascular Surgery, Georgetown University Hospital, MedStar Health, Washington, DC, USA
* Corresponding author. 106th Irving Street Northwest, Suite 3150, POB N, Washington, DC 20010.
E-mail address: Javairiah.Fatima@MedStar.net

Surg Clin N Am 103 (2023) 595–614
https://doi.org/10.1016/j.suc.2023.05.001 surgical.theclinics.com

surgical repair is preferably limited to high-volume centers and is reserved for aneurysms with complex anatomy that have a high risk of failure with endovascular intervention.

EPIDEMIOLOGY AND NATURAL HISTORY OF ABDOMINAL AORTIC ANEURYSM

The true incidence of abdominal aortic aneurysms is uncertain, but detection continues to increase in developed countries as the population ages and more screening programs become established.[1] Increased age, tobacco smoking, biological male gender, and family history remain the most important risk factors for the development of abdominal aortic aneurysm. Additional risk factors include hypertension, dyslipidemia, coronary artery disease, and connective tissue disease. Diabetes mellitus is somewhat protective for uncertain reasons.[6] The overall incidence of abdominal aortic aneurysm in patients below 55 years of age is very low. From 55 to 64 years, there is a roughly 1% prevalence and this increases by 3% per decade thereafter.[7–9] Each year, approximately 10,000 deaths occur in the United States due to ruptured abdominal aortic aneurysm.[10] Tobacco use serves as the primary modifiable risk factor linked to the development and rupture of abdominal aortic aneurysms. One in four patients enrolled in screening programs are active smokers and these patients are seven times more likely to develop an aneurysm when compared to non-smokers. The association between smoking and aneurysm development is particularly strong for women.[11] The duration of smoking rather than the total number of cigarettes smoked appears to determine risk.[12] Smoking cessation is associated with a significant risk reduction and after 25 years of cessation the risk is similar to that observed in non-smokers.[13] Caucasian race is the most frequently associated with the development of AAA, whereas there is decreased risk among Hispanics (OR 0.69), African Americans (OR 0.72), and Asians (OR 0.72).[14] First-degree family history of an abdominal aortic aneurysm increases risk by two-fold and is not gender-specific.[15] Lastly, the finding of peripheral arterial aneurysms confers increased risk and necessitates prompt evaluation of the abdominal aorta for aneurysmal disease.[16] Once an aneurysm develops, dilation continues slowly, albeit progressively. In accordance with the law of Laplace, smaller aneurysms increase at a slower rate, whereas larger aneurysms in the 4–5 cm range expand faster, at an approximate rate of 0.3–0.5 cm/year[17] The rate of expansion is, however, nonlinear in that most will progress but some do not, making individual predictions difficult.

SCREENING AND SURVEILLANCE OF ABDOMINAL AORTIC ANEURYSM

Population-based screening reduces abdominal aortic aneurysm-related mortality and rates of rupture. Screening is supported by data from four population-based randomized controlled trials (**Table 1**). Targeted screening programs have proven to be both cost-effective and logistically simple to implement.[22] The modality of choice for aneurysm screening is ultrasound, which is inexpensive and non-invasive and has sensitivity and specificity of 94% and 98% for the detection of aortic aneurysm.[23] The benefit of a single ultrasound screen extends out to 15 years follow-up.[24] United States Preventive Services Task Force (USPSTF) recommends screening be offered to men who have ever smoked, between the ages of 65 and 75 years.[25] Medicare beneficiaries who meet these inclusion criteria qualify for screening under the SAAVE act passed in 2007. Societal recommendations have somewhat broader inclusion recommendations and are summarized in **Table 2**. There is growing support for the expansion of screening programs beyond those outlined above.[14,30] One rationale is that screening trials proved effective during the era of open surgery when perioperative

Table 1
Landmark trials of population-based screening for AAA

Randomized Trial	Journal/ Year	Population Screened (n = no. Screened)	Positive Screens (n = Aneurysms)	Reduction from Aneurysm Mortality
MASS[18]	Lancet 2002	Men 65–74 yrs (n = 27,147)	4.9% (n = 1333)	Risk reduction 42% (95% CI 22–58)
Chichester[19]	Br J Surg 1995	Men and women 65–80 yrs (n = 5,394)	4.0% (n = 218)	Risk reduction 42% (95% CI NR)
Viborg[20]	BMJ 2005	Men 64–73 yrs (n = 6,333)	4.0% (n = 191)	Risk reduction 67% (95% CI 29–84)
Western Australia[21]	BMJ 2004	Men 65–79 yrs (n = 12,203)	7.2% (n = 875)	Risk ratio 0.61 (95% CI 0.33–1.11)

mortality rates were significantly higher than those seen in contemporary endovascular practice. Another driving force for expansion is the lower rates of repair in females and non-smokers.[31] The lower prevalence of comorbidities such as smoking and more advanced age at presentation are possible explanations why screening women under similar inclusion parameters to men has not proven beneficial.[11,32] Future directions aim at understanding which groups beyond the traditional scope need to be targeted for expansion of screening. Over 90% of aneurysms detected through screening will be below the size threshold for repair. These aneurysms instead undergo periodic ultrasound surveillance. Recommended surveillance intervals vary slightly among societal guidelines. In general, aneurysms under 4 cm should undergo imaging at 3-year intervals and those between 4.0 cm and 4.9 cm should be imaged annually. Once at the 5 cm mark, screening should be performed every 3 to 6 months until the threshold for repair is met. In women, 6-monthly screening may begin at 4.5 cm. Adherence to surveillance guidelines is fundamental for screening programs to be effective and has been reported to be as low as 65%.[19]

THRESHOLDS FOR ELECTIVE REPAIR OF ABDOMINAL AORTIC ANEURYSM

The decision to intervene should consider the likelihood of rupture, the individualized risks associated with repair and life expectancy of the patient. The diameter of an aneurysm is a surrogate for radial wall stress and remains the most accurate predictor of rupture. The cumulative five-year rate of rupture beyond 5 cm is approximately 30 to 40%, while smaller aneurysms in the sub-5 cm range have a much lower rupture rate at 1 to 5% per year.[33–35] Repair should be considered for small aneurysms that demonstrate expansion >0.5 cm in a 6 mth period. Such rapid growth is often representative of an unstable aortic wall. Heavy smokers and patients with poorly controlled hypertension are more likely to demonstrate rapid growth whereas this phenomenon is seen less often in diabetic patients.[36] The most widely accepted size threshold for intervention is $\geq$ 5.5 cm in men and between 5.0 cm and 5.4 cm for women.[4,26,27,37] Symptomatic aneurysms, typically in the form of abdominal or back pain, may be a sign of impending rupture and warrant urgent repair even if below the recommended size threshold.

REPAIR OF SMALL ABDOMINAL AORTIC ANEURYSMS

Repair of small aneurysms has been evaluated in several randomized trials. The UKSAT study randomized patients to either early open repair in the 4.0 cm to 5.5 cm range, or continued surveillance and open repair at the >5.5 cm threshold.

Table 2
Summary of AAA screening recommendations from major cardiovascular societies

Society	Year	Summary of AAA Screening Guidance
Society for Vascular Surgery (SVS)[4]	2018	Men or women aged 65–75 yrs with history of tobacco use First-degree relatives of patient with AAA aged 65–75 yrs or >75rs and in good health Men or women >75 yrs in good health if not screened prior
European Society for Vascular Surgery (ESVS)[26]	2019	Men aged 65 yrs Men and women aged ≥50 yrs who have first deg relative with AAA Population based screening for women not recommended
American College of Cardiology (ACC)/American Heart Association (AHA)[27]	2005	Men aged 65–75 yrs with history of smoking Men 60 years of age who are either siblings or offspring of patients with AAAs
American College of Preventive Medicine (ACPM)[28]	2011	Men aged 65–75 yrs with history of smoking
Canadian Society for Vascular Surgery (CSVS)[29]	2007	Men aged 65–75 yrs of age who are candidates for surgery, including those without significant smoking history

There was no difference in long-term mortality and the early surgery group experienced a perioperative mortality of 5.8%.[38] These findings were subsequently confirmed by the ADAM study, even at a lower operative mortality rate of 2%.[39] The CESAR trial compared surveillance to repair at 4.1 cm to 5.4 cm, but used EVAR instead of open repair.[40] There was no difference in all-cause mortality at 54 months; 14.5% in the EVAR group and 10.1% in the surveillance group, $p = 0.6$. Rates of aneurysm-related mortality, aneurysm rupture, and major morbidity were similar in both groups. The PIVOTAL trial enrolled patients with 4 cm to 5 cm abdominal aortic aneurysms and similarly found no mortality benefit of early endovascular intervention. Furthermore, across each of the trials, the endovascular repair of small aneurysms now mandated lifelong surveillance and was associated with reintervention. Beyond the outcomes of survival, repair of aneurysms in the small range does not offer improve quality of life or reduce costs.[41]

HISTORICAL PERSPECTIVES ON ENDOVASCULAR ANEURYSM REPAIR

The high perioperative morbidity and mortality associated with open aneurysm repair anticipated an early alternative. Suggestions of an intraluminal repair, thereby negating the need to open the abdomen, surfaced by the mid-1980s.[42] These initiatives in the aorta were based on an existing knowledge of stents in animals, used either alone or in combination with a polyester lining.[43,44] In 1987, Nocolai Volodos and his team in Soviet Ukraine performed the first EVAR in a human patient, successfully repairing a traumatic thoracic aortic pseudoaneurysm.[45] Four years later in Buenos Aires, Juan Parodi together with Julio Palmaz documented five successful infrarenal aneurysm repairs by retrograde deployment of a stent-anchored straight tubular endograft.[46] Thereafter, numerous similar successes were documented in the United States, Europe and Australia. Operator-designed endografts with monoiliac and bi-iliac configurations were introduced by the mid-1990s and commercial input followed as the industry recognized the potential of the new technology. Concurrent with the refinement of infrarenal EVAR, the treatment of more complex aortic anatomy began, inspired by fenestrated concepts in Australia.[47] By 1999, the Food and Drug Administration (FDA) approved the first two stent-grafts for commercial use; the Medtronic AneuRx and the Guidant Ancure. With the turn of the century, multiple manufacturers received approvals and there were significant improvements in delivery systems, fixation apparatus, and conformability, while simultaneously allowing for smaller sheath sizes. In the year 2000, more than half of all aneurysm repairs in the United States were performed with an endograft.[48] A fenestrated endograft received FDA approval in 2012 and an iliac-branched device in 2016. Today, EVAR accounts for almost 85% of aneurysm repairs in the United States and approximately 25,000 procedures are performed annually.[49] Seven devices are available for off-the-shelf repair of infrarenal aneurysmal disease, each of which has undergone several iterations since its initial approval, with ongoing research sustaining continued evolution.

EVIDENCE FOR ENDOVASCULAR REPAIR OF ABDOMINAL AORTIC ANEURYSMS

EVAR-1 was the first randomized prospective trial to demonstrate a reduction in perioperative mortality in patients undergoing EVAR comparted to traditional open surgical repair.[50] This UK-based multicenter trial enrolled 1082 patients who were deemed fit for open repair and demonstrated a reduction in aneurysm-related deaths in patients who underwent EVAR (4% vs 7%, HR 0.55), but no difference in all-cause mortality. The DREAM trial in the Netherlands was a smaller trial with 345 patients and used a threshold of 5.0 cm for intervention.[51] The trial reported a reduction in operative mortality in patients with EVAR (1.2% vs 4.6%) and reduced major perioperative complications (4.7% vs 9.8%). The OPEN trial similarly compared open repair and EVAR in 881 patients in the Veterans Affairs healthcare system in the United States.[52] Perioperative mortality was significantly lower for endovascular repair (0.5% vs 3%) and EVAR was associated with improvement in other hospital metrics including procedural time, ICU stay and total length of hospitalization. The ACE trial in France compared EVAR to open surgery in a much lower risk group of patients and found that in these patients there was no mortality benefit, and those who received endovascular treatment required more reinterventions.[53] The EVAR-2 trial enrolled patients who were unfit for open repair, and randomized to either EVAR or no intervention.[54] Survival was not improved with EVAR and was associated with a considerable 30-day mortality. Mortality benefits of endovascular repair were most profound in the early postoperative period and in all three of the EVAR-1, DREAM, and OVER trials, this survival advantage was lost in the mid-term. In EVAR-1, after 2 years there was no difference in all-cause mortality, at least in part because of fatal endograft ruptures, but also from more cardiovascular deaths in those undergoing endovascular repair.[55] In the DREAM trial, the perioperative survival benefit of EVAR did not extend beyond one year after randomization, but there was sustained benefit in aorta-related mortality. Similarly to EVAR-1, rates of reintervention were higher. Noninferiority of endovascular repair in the DREAM trial was maintained to six years and twelve years from randomization.[56,57] In the OVER trial, the benefit of endovascular repair was sustained at two three years, and after a mean follow-up of 5.2 years the survival was similar.[58]

EVAR DEVICE CHARACTERISTICS AND SELECTION

There are no prospective data comparing outcomes among EVAR devices and all are cost-comparable. Operator device selection is based upon knowledge of device characteristics and the unique anatomy of the aneurysm being repaired. While each EVAR device has features that may either limit or support its use, there are several overarching principles for device selection. First, all devices will perform adequately in patients with straightforward anatomy i.e., disease-free access vessels of adequate caliber, minimal iliac tortuosity and healthy, minimally-angulated aortic neck without prohibitive calcification or mural thrombus. Second, the user should stay within the confines of the instructions for use (IFU).[59,60] Nonadherence, in particular with respect to the proximal neck IFU, bears relationship to the development of 1a endoleak, reinterventions, EVAR failure, and possibly aortic-related mortality.[61] Third, operative familiarity with all available FDA devices will facilitate selection decisions based on technical nuances of the individual devices that cannot be otherwise obtained from the literature. Evolutionary timelines, anatomic requirements, and material characteristics for each of the seven FDA-approved EVAR devices are presented in **Tables 3–5**.

PREPROCEDURAL PLANNING AND EVAR DEVICE SIZING

Endovascular repair begins with patient selection, risk stratification, and additional pre-operative planning through device sizing. Computed tomography angiography is the cornerstone of imaging, ideally with 1 mm cuts in the axial plane. Post-processing three-dimensional reconstructions with centerline technology facilitate more precise measurements for sizing and planning. The overarching principle of sizing is to stay within the anatomic constraints of the IFU for the selected device. Review of the CT angiogram begins with a close examination of the seal zone. The presence of an angulated or conical neck, mural thrombus, or a high calcium burden is considered hostile neck features that may compromise proximal seal and result in EVAR failure. Healthy-appearing, parallel aortic neck is necessary to avoid failure of proximal seal. Ten to 20% oversizing to the aortic neck diameter is recommended. The neck diameter is measured from adventitia to adventitia for all endografts, except the Excluder C3 and Excluder Conformable (W.L. Gore) which is measured from intima to intima. Neck angulation in the craniocaudal axis is calculated to assist with the orthogonal orientation of the C-arm. Length measurements are then obtained from the lowest renal to the hypogastric arteries, ideally on using the centerline of flow to minimize inaccuracies in measurements. These measurements may also be obtained with a marker catheter during angiography. Next, the common iliac diameters are measured, and limbs are sized 10 to 20% beyond the diameters of the distal fixation sites. The common femoral arteries are evaluated for percutaneous access suitability, along with external iliac artery diameters to ensure sheath sizes of the selected device will be easily accommodated.

TECHNICAL APPROACH TO ENDOVASCULAR AORTIC ANEURYSM REPAIR

Endovascular aneurysm repair is most effectively performed in an operating room with fixed-mount imaging supported by staff trained in both open and endovascular techniques. The procedure can be performed under local anesthesia with conscious sedation or general anesthesia. The choice of anesthesia is tailored to the individual case at hand. Pedal Doppler signals are marked and recorded. The patient is then positioned supine and draped from nipples to knees for optimal exposure. Endovascular operators must be mindful of approaches to reduce radiation exposure throughout the

Table 3
FDA-approved EVAR devices available for use in the United States

Device	Excluder C3	Zenith Flex	AFX2	Endurant II	Alto	Excluder Conformable	Treo
Manufacturer	W. L. Gore & Associates	Cook Inc	Endologix LLC	Medtronic	Endologix LLC	W. L. Gore & Associates	Terumo Aortic
Device Name as Marketed	Excluder C3 conformable endoprosthesis	Zenith Flex AAA Endovascular Graft	AFX2 Endovascular AAA System	Endurant II Stent Graft System	Alto Abdominal Stent Graft System	Excluder Conformable AAA Endoprosthesis	TREO Abdominal Stent Graft System
Initial FDA Premarket Approval	2002	2003	2004	2010	2012	2020	2020
Current Iteration and year of Approval	3rd gen 2010	2nd gen 2008	3rd gen 2016	3rd gen 2017	4th gen 2020	1st gen 2020	1st gen 2020
Notable Revisions to prior Generations	Reduction in graft porosity to reduce type IV endoleak	Increase in conformability of iliac limbs	Change in fabric to reduce tears causing type III endoleak	Anchoring pins added to suprarenal fixation	Revision of filling channel to mitigate polymer leaks	Evolution of Excluder C3; reconstrainable with improved angulation control	NA

Table 4
Sizing requirements per the instructions for use (IFU) for FDA-approved EVAR devices

Device	Excluder C3	Zenith Flex	AFX2	Endurant II	Alto	Excluder Conformable	Treo
Proximal Neck Length	≥15 mm	≥15 mm	≥15 mm	≥10 mm	≥7 mm	≥10 mm, ≥ 15 mm	15 mm
Range of Neck Diameters	19–32 mm	18–32 mm	18–32 mm	19–32 mm	16–30 mm	16–32 mm	17–32 mm
Infrarenal Neck Angulation	≤60°	≤60°	≤60°	≤60°	≤60°	≤60° ≤90°	≤60°
Suprarenal Neck Angulation	N/A	≤45°	N/A	N/A	N/A	N/A	≤45°
Distal Fixation Length	≥10 mm	≥10 mm	≥15 mm	≥15 mm	≥10 mm	≥10 mm	≥10 mm
Range of Iliac Diameters	8–13.5 mm	7.5–20 mm	10–23 mm	8–25 mm	8–25 mm	8–25 mm	8–20 mm

Table 5
Material components and characteristics of FDA-approved EVAR devices

Device	Excluder C3	Zenith Flex	AFX2	Endurant II	Alto	Excluder Conformable	Treo
Graft Material	ePTFE and FEP	Woven polyester	Multilayered ePTFE	High-density women polyester	ePTFE	ePTFE	Woven polyester
Stent Material	Nitinol	Stainless steel main body, nitinol limbs	Cobalt chromium alloy	Nitinol	Nitinol	Nitinol	Nitinol
Fixation Features	Infrarenal fixation with barbs	Suprarenal fixation with barbs	Anatomical fixation on bifurcation and suprarenal fixation with barbs	Suprarenal fixation with barbs	Suprarenal fixation with barbs and polymer rings	Infrarenal fixation with barbs	Suprarenal and infrarenal
Main Body Access Profile	18–20Fr	18–22 Fr	17 Fr	18–20 Fr	15 Fr	15–18 Fr	18–19 Fr
Delivery System	Sheath required	Ensheathed	Sheath required	Sheathless	Ensheathed	Sheath required	Sheathless

procedure.[62] Bilateral percutaneous femoral access is achieved using a combination of ultrasound and fluoroscopic guidance. Using the preclose technique, suture-mediated closure devices are deployed. In the setting of diseased access vessels, open femoral exposure remains preferable to percutaneous access. The patient is heparinized and introducer sheaths are exchanged for large-bore sheaths. The device is introduced in the aorta over a super-stiff wire. Gantry angles are adjusted to remove parallax errors. Aortography is performed under apnea, and the inferior margin of the renal arteries is marked with roadmap guidance. Intravascular ultrasound and image-fusion software are useful adjuncts in device positioning and reduce radiation exposure and contrast use. The device is repositioned and consideration is given to gate orientation in an effort to facilitate favorable cannulation, including the need for limb crossing. The device is then deployed with the top of the fabric at or marginally below (eg, 1 to 2 mm) the lowest renal artery, based on the IFU for each device. Next, the gate is cannulated. Confirmatory maneuvers are mandatory to ensure wire position is truly within the graft and avoid potential deployment of the limb outside the gate and within the aneurysm sac. An oblique pelvic arteriogram is performed to mark the hypogastric artery before the deployment of the contralateral iliac limb. The remainder of the main body and ipsilateral limb are then deployed in a similar fashion. Seal zones and modular junctions are apposed with a compliant molding balloon. Stiff wires are withdrawn to allow the device to sit naturally in the aorta and completion arteriography is performed to evaluate for endoleak. Sheaths are removed and percutaneous sutures are fastened to close the arteriotomy. Soft tissues and skin incisions are closed. Technical approaches for complex endovascular repairs are beyond the scope of this review and have been expertly addressed elsewhere.[63]

POST-EVAR IMAGING SURVEILLANCE

Postoperative surveillance facilitates the detection of endoleaks, sac expansion, limb kinks, stent fractures, and graft migration, each of which must be addressed to maintain long-term success of the endovascular repair. Surveillance protocols have generally followed early randomized trials and used computed tomography angiography at 1-month, 6-month, and 12-month intervals, followed by yearly thereafter. The preferred protocol is a triphasic scan in the unenhanced, arterial, and delayed venous phases. The major downsides of computed tomography are contrast nephrotoxicity and the increased cost and radiation exposure when compared to ultrasound. Currently, data demonstrates that adherence to surveillance protocols is highly variable and less than half of patients will follow published guidance.[64] Furthermore, when guidelines are followed it results in more interventions but does not improve mortality.[65] As a result, there has been a movement toward less-intense post-EVAR surveillance, or perhaps even no surveillance in select patients.[66] One of the first modifications to traditional surveillance included the omission of the first 6-month scan if the 1-month CT angiogram was satisfactory, and this has now been widely incorporated into guidelines.[4,67] If the 12-month computed tomography angiogram demonstrates no endoleak, with either a stable sac size or sac regression, the patient may be enrolled in yearly duplex ultrasound instead of computed tomography. Limitations of ultrasound include operator dependence and poor visualization of the aorta in patients with large body habitus or overlying bowel gas. Duplex can accurately determine sac size and detect endoleaks, at a sensitivity similar or better than CT when enhanced with microbubbles.[68,69] Indeed, rigorous contrast-enhanced ultrasonography protocols will suffice in selected patients as the sole imaging modality post-EVAR.

CLASSIFICATION AND MANAGEMENT OF ENDOLEAKS

An endoleak is defined by persistent flow into an aneurysm sac after repair with a stent-graft device, leading to pressurization and potential expansion of the aneurysm sac, **Fig. 1**.[70] Late complications of EVAR are largely attributable to endoleaks, and the rates of reintervention are relatively high at 10–15%.[71]

Type I endoleaks refer to persistent blood flow around the graft at the proximal aortic neck seal (Type IA) or distally at the iliac seal zones (Type IB). Loss of proximal seal confers markedly elevated sac pressures and increased risk of rupture, and every attempt should be made to address type 1 endoleaks identified intraoperatively.[72] If there is sufficient length of uncovered aortic neck between the graft fabric and the lowest renal, proximal cuff extension is the most straightforward maneuver. Beyond this, proximal extension will involve the management of renal and visceral vessels, including fenestrated or branch endografts or open repair with explant.[73] With proximal leaks from inadequate radial apposition, for example in the setting of thrombus or calcium, buttressing with a Palmaz bare metal stent, can be effective.[74] Type 1 A endoleaks detected later are often due to caudal device migration in the setting of progressive natural or degenerative dilation of the aortic neck. Distal type 1b endoleaks are treated with the extension of the iliac limb and commonly this involves coil embolization and coverage of the hypogastric artery, or ideally extension with an iliac branched device in order to preserve hypogastric flow. If endovascular approaches do not offer resolve, consideration must be given to open conversion.

Type II endoleaks are caused by back-bleeding from vessels that originate from the wall of the sealed aorta, such as the inferior mesenteric artery, lumbar arteries, or middle sacral artery. Type II endoleaks are present in 10–20% of EVARs on post-operative computed tomography scans and follow an overall benign course when compared to type I and type III endoleaks.[75] The main factors consistently associated with type II endoleaks include a patent inferior mesenteric artery, increased number of lumbar arteries and maximum aneurysm diameter. With these factors accounted for, the use of anticoagulation or antiplatelets does not confer added risk of a type II endoleak.[76] The rates of rupture are very low at approximately 1%, but as many as one-third occur without documented sac expansion. Resolution of the endoleak without intervention

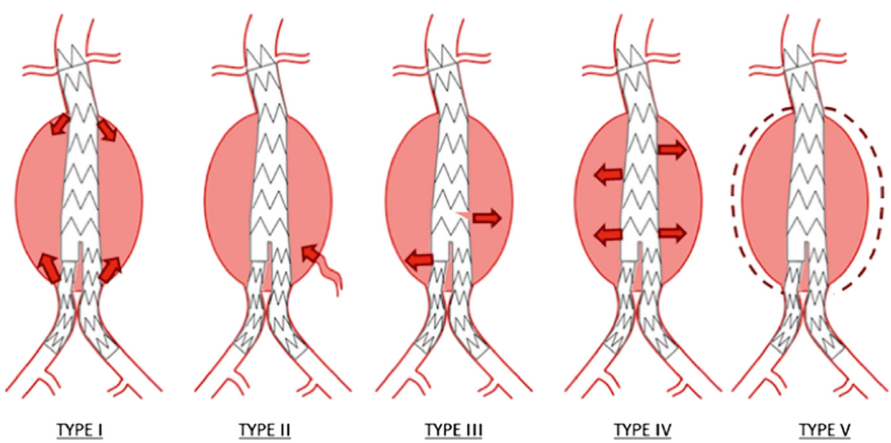

Fig. 1. Endoleak classification system. (A. England, R. McWilliams Endovascular Aortic Aneurysm Repair (EVAR) Ulster Med J, 82 (1) (2013), pp. 3-10.)

occurs in 35% of cases.[77] The current approach is to intervene when there is documented sac enlargement. The Society of Vascular Surgery recommends intervention for aneurysm sac expansion of ≥5 mm, whereas the European Society of Vascular Surgery guidelines recommend treatment for sac expansion of ≥10 mm, both based on level C evidence.[4,26] One must be mindful that sac enlargement thought to be secondary to type II endoleaks may be the harbinger of an obscure type 1a endoleak and careful review is warranted. Embolization of the inflow and outflow vessels has a high success rate and may be performed with liquid embolic agents or coils. Routes to access the aneurysm sac include translumbar, transcaval or via the SMA-IMA arcade.

Type III endoleaks result from component separation in modular grafts (Type IIIa) or from tears in the graft fabric (Type IIIb). Component separation is caused by insufficient overlap between endograft modules, often precipitated by geometric changes in the aneurysm over time or from device migration. Fabric tears may be caused by stent fractures, calcific lesions, or erosion points. Most type III leaks therefore develop months to years beyond the index procedure.[78] The overall incidence is low across all devices, ranging from 3 to 5% in registry data.[79] Since type III endoleaks result in direct flow into the aortic sac, rates of secondary rupture are high and urgent repair is indicated.

Type IV endoleaks are caused by leakage into the sac across blood-porous fabric, classically seen as a blush across the fabric. It is no longer seen with newer-generation endografts. Type IV endoleaks self-resolve as the graft interstices thrombose and normal coagulation pathways resume.

Type V endoleak is somewhat of a misnomer and refers to continued sac expansion with increased pressures, but without demonstrable endoleak. The phenomenon may also be termed “endotension.”The mainstay of treatment remains explant and open repair, although relining may be considered for higher-risk patients.

ENDOVASCULAR REPAIR OF RUPTURED ABDOMINAL AORTIC ANEURYSMS

Ruptured abdominal aortic aneurysm is a catastrophic event with an overall mortality of 80–90%.[80] Of those that survive until the point of hospital contact, the 30-day mortality is approximately 50%.[3] Mortality from ruptured AAA follows bimodal distribution. Death within the first 48 hrs is typically from hemorrhagic shock. Death beyond this immediate period, often in spite of successful repair, is related to sequelae of the initial critical illness compounded by comorbid conditions.[81] Several retrospective studies have noted a trend toward increased survival for ruptured abdominal aortic aneurysms in contemporary practice.[82–84] Arguably, the most major advancement is the use of EVAR for ruptured aneurysms. Endovascular therapy offers rapid balloon control of the proximal aorta, an overall shorter operative time and the ability for the procedure to be done under local anesthesia. This prompted several large single-center series, national registry, studies and randomized trials to evaluate the benefit of EVAR over open surgical repair for ruptured abdominal aortic aneurysm. US Medicare data derived from approximately 1100 propensity-matched pairs reported a significantly lower mortality with EVAR compared to open surgery (34% vs 48%, $p < 0.001$).[85] Similar findings were seen from an analysis of 21,206 ruptured abdominal aortic aneurysms across the United States as part of the Nationwide Inpatient Sample (NIS) database.[86] EVAR demonstrated a benefit over open repair for both mortality (OR 0.54) and major complications (OR 0.49). One of the major limitations of the aforementioned observational data is selection bias, whereby patients plausibly received an open operation because of highly unfavorable EVAR anatomy. The bias lies in the fact that patients with challenging landing zones will often require a more complex open operation, for example necessitating suprarenal exposure and clamping. That the mortality benefit of EVAR was not reproduced under the rigor randomized trial was unsurprising. However, close attention to each of these trials offers important insights into understanding the role of EVAR for rAAA. The AJAX trial, based across three centers in Amsterdam, randomized 116 ruptured infrarenal AAAs that were fit for either type of repair. The primary endpoint was a composite of death and major complications, which occurred in 42% of the EVAR group and 47% of the open surgical repair group (95% CI -13% – 23%).[87] There was a significant reduction in renal failure in the EVAR group (ARR 20%). One key limitation of the AJAX trial was the exclusion of patients hemodynamically unfit for CTA. The IMPROVE trial included 316 patients which were randomized into open and endovascular groups after computed tomography imaging was obtained.[88] The design allowed crossover to the open group if aortic morphology was unsuitable for endovascular repair, closely mirroring the reality of clinical practice. The 30-day mortality was not significantly different between groups (35.4% vs 37.4%, $p = 0.62$), although in the subset of women a significant mortality reduction was observed. Important secondary outcomes included a shorter ICU length of stay and higher rates of discharge to home in the endovascular group. Of note, the endovascular approach was not more cost-effective. The French ECAR trial randomized hemodynamically unstable ruptured AAAs that were suitable to both open and endovascular repair.[89] Mortality at 30 days and 1 year were not different between the groups (18% in the endovascular group vs. 24% in the open group at 30 days, and 30% vs. 35% respectively at 1 year). Pulmonary complications, need for transfusion and ICU length of stay were lower in the endovascular group.

CONTEMPORARY OUTCOMES OF OPEN ABDOMINAL AORTIC ANEURYSM REPAIR

Open surgical repair of abdominal aortic aneurysms became a routine part of vascular practice by the 1970s and has undergone continued technical refinement and evolution. It is the gold standard to which EVAR is compared and remains one of the most academically scrutinized surgical procedures. The leading cause of mortality after open aneurysm repair is related to comorbid coronary artery disease and efforts to stratify and reduce this risk are paramount.[90] Management of pre-operative obstructive lung disease, including smoking cessation, and optimization of renal function also significantly reduces risk.[91–93] Historically, the post-operative mortality of open abdominal aortic aneurysm repair has been quoted at 5–8%.[94,95] These figures continue to improve and contemporary mortality, supported by figures from the UK National Vascular and US Nationwide Inpatient Sample databases are best estimated in the 3 to 5% range.[96,97] The open arms of DREAM, OVER and EVAR-1 also reports similar mortality.[50–52] Data from tertiary-level aortic centers, which largely include high-risk patients have consistently reported a $\leq$3% mortality.[98,99] These data support the practice whereby younger patients with low anesthetic risk and few comorbidities are offered open repair as first-line treatment for infrarenal aneurysms, provided the procedure is done in a center of expertise. Contemporary open outcomes occur in an era of unprecedented endovascular innovation whereby the open solution inherently involves a more challenging operation such as suprarenal clamping, explant of a failed endovascular device, or dealing with concomitant aortoiliac occlusive disease.

SUMMARY

Abdominal aortic aneurysms continue to pose considerable challenges in the daily practice of vascular surgery. Ruptured aortic aneurysms are almost uniformly fatal, irrespective of the type of repair used, and mortality reduction is achieved by repair in the asymptomatic stage. Advancements in endovascular technology have significantly reduced perioperative mortality and remain noninferior to open repair in the mid-term, but with higher rates of reintervention. Longer term trial data are not available for endovascular repair but the durability of an open repair is definite. An intricate understanding of the natural history of the disease and the patient at hand is imperative when deciding on the best modality of treatment. Each of the endovascular devices available is comparable in performance and selection is based on the anatomic features of the aneurysm. Device failure is frequent when the devices are used outside of the instructions for use and this has been a major failing of the endovascular revolution to date.

CLINICS CARE POINTS

- Abdominal aortic aneurysms are found in 5% of men over the age of 65 years.
- Screening programs reduce aneurysm-related mortality in selected populations.
- Endovascular repair is the primary modality of repair in contemporary practice.
- Endovascular repair is limited in anatomically more complex aneurysms.
- Open surgery may be offered as first-line treatment in younger patients with low anesthetic risk.
- Ruptured aneurysms continue to have very high mortality irrespecitve of approach used.

REFERENCES

1. Sampson UK, Norman PE, Fowkes FG, et al. Estimation of global and regional incidence and prevalence of abdominal aortic aneurysms 1990 to 2010. Glob Heart 2014;9(1):159–70.
2. Scott RA, Ashton HA, Kay DN. Abdominal aortic aneurysm in 4237 screened patients: prevalence, development and management over 6 years. Br J Surg 1991; 78(9):1122–5.
3. Bown MJ, Sutton AJ, Bell PR, et al. A meta-analysis of 50 years of ruptured abdominal aortic aneurysm repair. Br J Surg 2002;89(6):714–30.
4. Chaikof EL, Dalman RL, Eskandari MK, et al. The Society for Vascular Surgery practice guidelines on the care of patients with an abdominal aortic aneurysm. J Vasc Surg 2018;67(1):2–77 e2.
5. Dua A, Kuy S, Lee CJ, et al. Epidemiology of aortic aneurysm repair in the United States from 2000 to 2010. J Vasc Surg 2014;59(6):1512–7.
6. De Rango P, Farchioni L, Fiorucci B, et al. Diabetes and abdominal aortic aneurysms. Eur J Vasc Endovasc Surg 2014;47(3):243–61.
7. Pleumeekers HJ, Hoes AW, van der Does E, et al. Aneurysms of the abdominal aorta in older adults. The Rotterdam Study. Am J Epidemiol 1995;142(12): 1291–9.
8. Singh K, Bonaa KH, Jacobsen BK, et al. Prevalence of and risk factors for abdominal aortic aneurysms in a population-based study : the Tromso Study. Am J Epidemiol 2001;154(3):236–44.
9. Lederle FA, Johnson GR, Wilson SE, et al. Prevalence and associations of abdominal aortic aneurysm detected through screening. Aneurysm Detection

and Management (ADAM) Veterans Affairs Cooperative Study Group. Ann Intern Med 1997;126(6):441–9.
10. Go AS, Mozaffarian D, Roger VL, et al. Heart disease and stroke statistics–2013 update: a report from the American Heart Association. Circulation 2013;127(1): e6–245.
11. Carter JL, Morris DR, Sherliker P, et al. Sex-specific associations of vascular risk factors with abdominal aortic aneurysm: findings from 1.5 million women and 0.8 million men in the United States and United Kingdom. J Am Heart Assoc 2020; 9(4):e014748.
12. Wilmink TB, Quick CR, Day NE. The association between cigarette smoking and abdominal aortic aneurysms. J Vasc Surg 1999;30(6):1099–105.
13. Aune D, Schlesinger S, Norat T, et al. Tobacco smoking and the risk of abdominal aortic aneurysm: a systematic review and meta-analysis of prospective studies. Sci Rep 2018;8(1):14786.
14. Kent KC, Zwolak RM, Egorova NN, et al. Analysis of risk factors for abdominal aortic aneurysm in a cohort of more than 3 million individuals. J Vasc Surg 2010;52(3):539–48.
15. Larsson E, Granath F, Swedenborg J, et al. A population-based case-control study of the familial risk of abdominal aortic aneurysm. J Vasc Surg 2009;49(1): 47–50 [discussion: 1].
16. Diwan A, Sarkar R, Stanley JC, et al. Incidence of femoral and popliteal artery aneurysms in patients with abdominal aortic aneurysms. J Vasc Surg 2000;31(5): 863–9.
17. Powell JT, Sweeting MJ, Brown LC, et al. Systematic review and meta-analysis of growth rates of small abdominal aortic aneurysms. Br J Surg 2011;98(5):609–18.
18. Ashton HA, Buxton MJ, Day NE, et al. The Multicentre Aneurysm Screening Study (MASS) into the effect of abdominal aortic aneurysm screening on mortality in men: a randomised controlled trial. Lancet 2002;360(9345):1531–9.
19. Scott RA, Wilson NM, Ashton HA, et al. Influence of screening on the incidence of ruptured abdominal aortic aneurysm: 5-year results of a randomized controlled study. Br J Surg 1995;82(8):1066–70.
20. Lindholt JS, Juul S, Fasting H, et al. Screening for abdominal aortic aneurysms: single centre randomised controlled trial. BMJ 2005;330(7494):750.
21. Norman PE, Jamrozik K, Lawrence-Brown MM, et al. Population based randomised controlled trial on impact of screening on mortality from abdominal aortic aneurysm. BMJ 2004;329(7477):1259.
22. Glover MJ, Kim LG, Sweeting MJ, et al. Cost-effectiveness of the national health service abdominal aortic aneurysm screening programme in England. Br J Surg 2014;101(8):976–82.
23. Guirguis-Blake JM, Beil TL, Senger CA, et al. Ultrasonography screening for abdominal aortic aneurysms: a systematic evidence review for the U.S. Preventive Services Task Force. Ann Intern Med 2014;160(5):321–9.
24. Ali MU, Fitzpatrick-Lewis D, Miller J, et al. Screening for abdominal aortic aneurysm in asymptomatic adults. J Vasc Surg 2016;64(6):1855–68.
25. Force USPST, Owens DK, Davidson KW, Krist AH, et al. Screening for abdominal aortic aneurysm: US preventive services task force recommendation statement. JAMA 2019;322(22):2211–8.
26. Wanhainen A, Verzini F, Van Herzeele I, et al. Editor's Choice - European Society for Vascular Surgery (ESVS) 2019 clinical practice guidelines on the management of abdominal aorto-iliac artery aneurysms. Eur J Vasc Endovasc Surg 2019; 57(1):8–93.

27. Hirsch AT, Haskal ZJ, Hertzer NR, et al. ACC/AHA 2005 Practice Guidelines for the management of patients with peripheral arterial disease (lower extremity, renal, mesenteric, and abdominal aortic): a collaborative report from the American association for vascular surgery/society for vascular surgery, society for cardiovascular angiography and interventions, society for vascular medicine and biology, society of interventional radiology, and the ACC/AHA task force on practice guidelines (writing committee to develop guidelines for the management of patients with peripheral arterial disease): endorsed by the American association of cardiovascular and pulmonary rehabilitation; national heart, lung, and blood institute; society for vascular nursing; transatlantic inter-society consensus; and vascular disease foundation. Circulation 2006;113(11):e463–654.
28. Lim LS, Haq N, Mahmood S, et al, Committee APP, American College of Preventive M. Atherosclerotic cardiovascular disease screening in adults: American college of preventive medicine position statement on preventive practice. Am J Prev Med 2011;40(3):381 e1–e10.
29. Mastracci TM, Cina CS, Canadian Society for Vascular S. Screening for abdominal aortic aneurysm in Canada: review and position statement of the Canadian Society for Vascular Surgery. J Vasc Surg 2007;45(6):1268–76.
30. O'Donnell TFX, Landon BE, Schermerhorn ML. The case for expanding abdominal aortic aneurysm screening. J Vasc Surg 2020;71(5):1809–12.
31. Levin SR, Farber A, Goodney PP, et al. The U.S. preventive services task force abdominal aortic aneurysm screening guidelines negligibly impacted repair rates in male never-smokers and female smokers. Ann Vasc Surg 2022;82:87–95.
32. Duncan A, Maslen C, Gibson C, et al. Ultrasound screening for abdominal aortic aneurysm in high-risk women. Br J Surg 2021;108(10):1192–8.
33. Johansson G, Nydahl S, Olofsson P, et al. Survival in patients with abdominal aortic aneurysms. Comparison between operative and nonoperative management. Eur J Vasc Surg 1990;4(5):497–502.
34. Lederle FA, Johnson GR, Wilson SE, et al. Rupture rate of large abdominal aortic aneurysms in patients refusing or unfit for elective repair. JAMA 2002;287(22):2968–72.
35. Nevitt MP, Ballard DJ, Hallett JW Jr. Prognosis of abdominal aortic aneurysms. A population-based study. N Engl J Med 1989;321(15):1009–14.
36. Brady AR, Thompson SG, Fowkes FG, et al, Participants UKSAT. Abdominal aortic aneurysm expansion: risk factors and time intervals for surveillance. Circulation 2004;110(1):16–21.
37. Chaikof EL, Brewster DC, Dalman RL, et al. SVS practice guidelines for the care of patients with an abdominal aortic aneurysm: executive summary. J Vasc Surg 2009;50(4):880–96.
38. Mortality results for randomised controlled trial of early elective surgery or ultrasonographic surveillance for small abdominal aortic aneurysms. The UK Small Aneurysm Trial Participants. Lancet 1998;352(9141):1649–55.
39. Lederle FA, Johnson GR, Wilson SE, et al. The aneurysm detection and management study screening program: validation cohort and final results. Aneurysm detection and management veterans affairs cooperative study investigators. Arch Intern Med 2000;160(10):1425–30.
40. Cao P, De Rango P, Verzini F, et al. Comparison of surveillance versus aortic endografting for small aneurysm repair (CAESAR): results from a randomised trial. Eur J Vasc Endovasc Surg 2011;41(1):13–25.
41. Filardo G, Powell JT, Martinez MA, et al. Surgery for small asymptomatic abdominal aortic aneurysms. Cochrane Database Syst Rev 2015;(2):CD001835.

42. Balko A, Piasecki GJ, Shah DM, et al. Transfemoral placement of intraluminal polyurethane prosthesis for abdominal aortic aneurysm. J Surg Res 1986;40(4):305–9.
43. Charnsangavej C, Wallace S, Wright KC, et al. Endovascular stent for use in aortic dissection: an in vitro experiment. Radiology 1985;157(2):323–4.
44. Wright KC, Wallace S, Charnsangavej C, et al. Percutaneous endovascular stents: an experimental evaluation. Radiology 1985;156(1):69–72.
45. Volodos NL, Karpovich IP, Shekhanin VE, et al. A case of distant transfemoral endoprosthesis of the thoracic artery using a self-fixing synthetic prosthesis in traumatic aneurysm. Grudn Khir 1988;(6):84–6.
46. Parodi JC, Palmaz JC, Barone HD. Transfemoral intraluminal graft implantation for abdominal aortic aneurysms. Ann Vasc Surg 1991;5(6):491–9.
47. Browne TF, Hartley D, Purchas S, et al. A fenestrated covered suprarenal aortic stent. Eur J Vasc Endovasc Surg 1999;18(5):445–9.
48. Anderson PL, Arons RR, Moskowitz AJ, et al. A statewide experience with endovascular abdominal aortic aneurysm repair: rapid diffusion with excellent early results. J Vasc Surg 2004;39(1):10–9.
49. Suckow BD, Goodney PP, Columbo JA, et al. National trends in open surgical, endovascular, and branched-fenestrated endovascular aortic aneurysm repair in Medicare patients. J Vasc Surg 2018;67(6):1690–1697 e1.
50. EVAR trial participants. Endovascular aneurysm repair versus open repair in patients with abdominal aortic aneurysm (EVAR trial 1): randomised controlled trial. Lancet 2005;365(9478):2179–86.
51. Prinssen M, Verhoeven EL, Buth J, et al. A randomized trial comparing conventional and endovascular repair of abdominal aortic aneurysms. N Engl J Med 2004;351(16):1607–18.
52. Lederle FA, Freischlag JA, Kyriakides TC, et al. Outcomes following endovascular vs open repair of abdominal aortic aneurysm: a randomized trial. JAMA 2009;302(14):1535–42.
53. Becquemin JP, Pillet JC, Lescalie F, et al. A randomized controlled trial of endovascular aneurysm repair versus open surgery for abdominal aortic aneurysms in low- to moderate-risk patients. J Vasc Surg 2011;53(5):1167–11673 e1.
54. EVAR trial participants. Endovascular aneurysm repair and outcome in patients unfit for open repair of abdominal aortic aneurysm (EVAR trial 2): randomised controlled trial. Lancet 2005;365(9478):2187–92.
55. United Kingdom ETI, Greenhalgh RM, Brown LC, et al. Endovascular versus open repair of abdominal aortic aneurysm. N Engl J Med 2010;362(20):1863–71.
56. De Bruin JL, Baas AF, Buth J, et al. Long-term outcome of open or endovascular repair of abdominal aortic aneurysm. N Engl J Med 2010;362(20):1881–9.
57. van Schaik TG, Yeung KK, Verhagen HJ, et al. Long-term survival and secondary procedures after open or endovascular repair of abdominal aortic aneurysms. J Vasc Surg 2017;66(5):1379–89.
58. Lederle FA, Freischlag JA, Kyriakides TC, et al. Long-term comparison of endovascular and open repair of abdominal aortic aneurysm. N Engl J Med 2012;367(21):1988–97.
59. Herman CR, Charbonneau P, Hongku K, et al. Any nonadherence to instructions for use predicts graft-related adverse events in patients undergoing elective endovascular aneurysm repair. J Vasc Surg 2018;67(1):126–33.
60. Abbruzzese TA, Kwolek CJ, Brewster DC, et al. Outcomes following endovascular abdominal aortic aneurysm repair (EVAR): an anatomic and device-specific analysis. J Vasc Surg 2008;48(1):19–28.

61. AbuRahma AF, Campbell J, Stone PA, et al. The correlation of aortic neck length to early and late outcomes in endovascular aneurysm repair patients. J Vasc Surg 2009;50(4):738–48.
62. Maurel B, Hertault A, Sobocinski J, et al. Techniques to reduce radiation and contrast volume during EVAR. J Cardiovasc Surg 2014;55(2 Suppl 1):123–31.
63. Oderich GS. Endovascular aortic repair: current techniques with fenestrated, Branched and parallel stent-grafts. 1e edition. Switzerland: Springer; 2017. p. 745.
64. Garg T, Baker LC, Mell MW. Adherence to postoperative surveillance guidelines after endovascular aortic aneurysm repair among Medicare beneficiaries. J Vasc Surg 2015;61(1):23–7.
65. Antoniou GA, Kontopodis N, Rogers SK, et al. Meta-analysis of compliance with endovascular aneurysm repair surveillance: The EVAR surveillance paradox. Eur J Vasc Endovasc Surg 2022;65(2):244–54.
66. Geraedts ACM, Mulay S, Vahl AC, et al. Editor's choice - post-operative surveillance and long term outcome after endovascular aortic aneurysm repair in patients with an initial post-operative computed tomography angiogram without abnormalities: the multicentre retrospective ODYSSEUS study. Eur J Vasc Endovasc Surg 2022;63(3):390–9.
67. Go MR, Barbato JE, Rhee RY, et al. What is the clinical utility of a 6-month computed tomography in the follow-up of endovascular aneurysm repair patients? J Vasc Surg 2008;47(6):1181–6 [discussion: 6-7].
68. Harky A, Zywicka E, Santoro G, et al. Is contrast-enhanced ultrasound (CEUS) superior to computed tomography angiography (CTA) in detection of endoleaks in post-EVAR patients? A systematic review and meta-analysis. J Ultrasound 2019;22(1):65–75.
69. Karaolanis GI, Antonopoulos CN, Georgakarakos E, et al. Colour duplex and/or contrast-enhanced ultrasound compared with computed tomography angiography for endoleak detection after endovascular abdominal aortic aneurysm repair: a systematic review and meta-analysis. J Clin Med 2022;11(13):3628.
70. Kassen T. Follow up CT angiography post EVAR: endoleaks detection, classification and manegement planning. Egypt J Radiol Nucl Med 2017;48(3):621–6.
71. Conrad MF, Adams AB, Guest JM, et al. Secondary intervention after endovascular abdominal aortic aneurysm repair. Ann Surg 2009;250(3):383–9.
72. Harris PL, Vallabhaneni SR, Desgranges P, et al. Incidence and risk factors of late rupture, conversion, and death after endovascular repair of infrarenal aortic aneurysms: the EUROSTAR experience. European Collaborators on Stent/graft techniques for aortic aneurysm repair. J Vasc Surg 2000;32(4):739–49.
73. Scali ST, Beck AW, Chang CK, et al. Defining risk and identifying predictors of mortality for open conversion after endovascular aortic aneurysm repair. J Vasc Surg 2016;63(4):873–881 e1.
74. Arthurs ZM, Lyden SP, Rajani RR, et al. Long-term outcomes of Palmaz stent placement for intraoperative type Ia endoleak during endovascular aneurysm repair. Ann Vasc Surg 2011;25(1):120–6.
75. Gelfand DV, White GH, Wilson SE. Clinical significance of type II endoleak after endovascular repair of abdominal aortic aneurysm. Ann Vasc Surg 2006;20(1):69–74.
76. Guo Q, Du X, Zhao J, et al. Prevalence and risk factors of type II endoleaks after endovascular aneurysm repair: a meta-analysis. PLoS One 2017;12(2):e0170600.

77. Sidloff DA, Stather PW, Choke E, et al. Type II endoleak after endovascular aneurysm repair. Br J Surg 2013;100(10):1262–70.
78. Maleux G, Poorteman L, Laenen A, et al. Incidence, etiology, and management of type III endoleak after endovascular aortic repair. J Vasc Surg 2017;66(4):1056–64.
79. Hobo R, Buth J, collaborators E. Secondary interventions following endovascular abdominal aortic aneurysm repair using current endografts. A EUROSTAR report. J Vasc Surg 2006;43(5):896–902.
80. Reimerink JJ, van der Laan MJ, Koelemay MJ, et al. Systematic review and meta-analysis of population-based mortality from ruptured abdominal aortic aneurysm. Br J Surg 2013;100(11):1405–13.
81. Reitz KM, Phillips AR, Tzeng E, et al. Characterization of immediate and early mortality after repair of ruptured abdominal aortic aneurysm. J Vasc Surg 2022;76(6):1578–87.e5.
82. Norman PE, Spilsbury K, Semmens JB. Falling rates of hospitalization and mortality from abdominal aortic aneurysms in Australia. J Vasc Surg 2011;53(2):274–7.
83. Sandiford P, Mosquera D, Bramley D. Trends in incidence and mortality from abdominal aortic aneurysm in New Zealand. Br J Surg 2011;98(5):645–51.
84. Bartek MA, Kessler LG, Talbott JM, et al. Washington State abdominal aortic aneurysm-related mortality shows a steady decline between 1996 and 2016. J Vasc Surg 2019;70(4):1115–22.
85. Edwards ST, Schermerhorn ML, O'Malley AJ, et al. Comparative effectiveness of endovascular versus open repair of ruptured abdominal aortic aneurysm in the Medicare population. J Vasc Surg 2014;59(3):575–82.
86. Park BD, Azefor N, Huang CC, et al. Trends in treatment of ruptured abdominal aortic aneurysm: impact of endovascular repair and implications for future care. J Am Coll Surg 2013;216(4):745–54 [discussion: 54-5].
87. Reimerink JJ, Hoornweg LL, Vahl AC, et al. Endovascular repair versus open repair of ruptured abdominal aortic aneurysms: a multicenter randomized controlled trial. Ann Surg 2013;258(2):248–56.
88. Investigators IT, Powell JT, Sweeting MJ, et al. Endovascular or open repair strategy for ruptured abdominal aortic aneurysm: 30 day outcomes from IMPROVE randomised trial. BMJ 2014;348:f7661.
89. Desgranges P, Kobeiter H, Katsahian S, et al. Editor's Choice - ECAR (Endovasculaire ou Chirurgie dans les Anevrysmes aorto-iliaques Rompus): a French Randomized Controlled Trial of Endovascular Versus Open Surgical Repair of Ruptured Aorto-iliac Aneurysms. Eur J Vasc Endovasc Surg 2015;50(3):303–10.
90. Fleisher LA, Beckman JA, Brown KA, et al. 2009 ACCF/AHA focused update on perioperative beta blockade incorporated into the ACC/AHA 2007 guidelines on perioperative cardiovascular evaluation and care for noncardiac surgery: a report of the American college of cardiology foundation/American heart association task force on practice guidelines. Circulation 2009;120(21):e169–276.
91. Upchurch GR Jr, Proctor MC, Henke PK, et al. Predictors of severe morbidity and death after elective abdominal aortic aneurysmectomy in patients with chronic obstructive pulmonary disease. J Vasc Surg 2003;37(3):594–9.
92. Lee SM, Takemoto S, Wallace AW. Association between withholding angiotensin receptor blockers in the early postoperative period and 30-day mortality: a cohort study of the veterans affairs healthcare system. Anesthesiology 2015;123(2):288–306.

93. Peterson L, Schweitzer G, Simone A, et al. The effect of smoking status on perioperative morbidity and mortality after open and endovascular abdominal aortic aneurysm repair. Ann Vasc Surg 2022;88:373–84.
94. Bradbury AW, Adam DJ, Makhdoomi KR, et al. A 21-year experience of abdominal aortic aneurysm operations in Edinburgh. Br J Surg 1998;85(5):645–7.
95. Heller JA, Weinberg A, Arons R, et al. Two decades of abdominal aortic aneurysm repair: have we made any progress? J Vasc Surg 2000;32(6):1091–100.
96. Hicks CW, Canner JK, Arhuidese I, et al. Comprehensive assessment of factors associated with in-hospital mortality after elective abdominal aortic aneurysm repair. JAMA Surg 2016;151(9):838–45.
97. Grant SW, Grayson AD, Mitchell DC, et al. Evaluation of five risk prediction models for elective abdominal aortic aneurysm repair using the UK National Vascular Database. Br J Surg 2012;99(5):673–9.
98. Hicks CW, Black JH 3rd, Arhuidese I, et al. Mortality variability after endovascular versus open abdominal aortic aneurysm repair in a large tertiary vascular center using a Medicare-derived risk prediction model. J Vasc Surg 2015;61(2):291–7.
99. Conrad MF, Crawford RS, Pedraza JD, et al. Long-term durability of open abdominal aortic aneurysm repair. J Vasc Surg 2007;46(4):669–75.